

# Ferran Alía Castillejos

Data science and physical engineering, UPC Barcelona

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## Now

In my fourth year at UPC in Barcelona, back from a summer simulating microscope optics with the Physics of Life group at TU Dresden. I want to work where physical modeling and machine learning meet.

## Experience

Since 2026

### Contributor, Towards Data Science

Long-form articles since March 2026, on simulation, high-performance computing and machine learning. Most of them come out of the projects below.

Jun – Aug 2026

### Research intern, Physics of Life, TU Dresden

Simulating electromagnetic propagation through specimen and objective, as virtual instrumentation for learned brightfield imaging. On the side, a GNN surrogate for multiple scattering between beads.

Jun – Aug 2025

### Machine learning intern, MLCode

Built an automated framework for evaluating and benchmarking retrieval-augmented generation systems.

Since 2022

### Volunteer mentor

Computer science and robotics workshops: Linux, 3D printing, programming.

2022 – 2023

### Robotics teacher, Punt Multimèdia

Taught Arduino, micro:bit, and 3D printing and modeling.

## Education

2023 – 2028

### BSc Data Science & Engineering, UPC

In the fourth year of five. 8.75/10 average across both degrees.

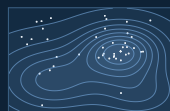
2023 – 2028

### BSc Physical Engineering, UPC

Taken in parallel with the data science degree rather than after it, on the CFIS dual-degree program.

## Selected projects

2026

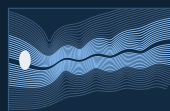


### Aerodynamic optimization of a Formula 1 front wing

A surrogate model searched front-wing geometries on a fixed budget of solver runs rather than calling CFD at every step. The wing came out about 15% better on lift-to-drag.

Python, OpenFOAM, SLURM, MareNostrum V

2026



### A physics-informed network for vortex shedding

A PINN with the Navier-Stokes residual inside its loss, trained against a high-fidelity OpenFOAM run, reproducing Kármán vortex shedding. A model fit to the data alone settles into a steady wake instead.

Python, PyTorch, OpenFOAM

2026



### A graph network for multiple scattering in microscopy

A GNN surrogate for the bead-bead coupling in Köhler-illuminated microscope images, fast enough to run in real time. Built at TU Dresden alongside the main internship project.

2026



### Persona adaptation in language models

A system prompt against three formats of fine-tuning data on Qwen3-4B-Instruct, all with LoRA at  $r=16$ , so the differences belong to the data.

Python, PyTorch, Transformers, LoRA

2025

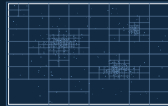


### Predicting fetal distress from cardiotocography

Built with Hospital Sant Joan de Déu in a team of nine. Separate encoders over CTG traces, tabular clinical context and ultrasound, at around 0.78 AUC.

Python, PyTorch

2025

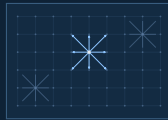


### Parallel N-body simulation

Direct  $O(n^2)$  summation and a Barnes-Hut  $O(n \log n)$  tree code in C++ with OpenMP. At  $N = 10,000$  direct summation ran about  $33\times$  faster on 112 threads than on one; the tree code never got past roughly  $7\times$ .

C++, OpenMP

2025



### Parallel lattice Boltzmann solver

A D2Q9 solver with BGK collision and a Zou-He inlet in around 200 lines of C++; flow past a cylinder on a  $400\times 400$  lattice. Memory-bound, so how you move the data dominates the scaling.

C++, OpenMP, MareNostrum 5

2025

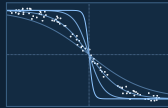


### Fixed-wing bird flight simulation

A 3D Navier-Stokes solver written from scratch in NumPy, run on a bird's wing modeled in Blender and checked against OpenFOAM's icoFoam.

Python, NumPy, OpenFOAM, Blender

2024



### Fermi-Dirac statistics by Monte Carlo

A Markov chain over the microstates of a fermion system, recovering the Fermi-Dirac occupancy numerically rather than analytically.

Python, NumPy

Figures are schematic illustrations of each problem, not results.

## Skills

|            |                                                                               |
|------------|-------------------------------------------------------------------------------|
| Languages  | Python, C++, R, Haskell, MATLAB                                               |
| Scientific | OpenFOAM, OpenMP, NumPy, PyTorch, Blender, CAD                                |
| Tools      | Linux, Git                                                                    |
| Methods    | CFD, HPC & SLURM, physics-informed ML, Monte Carlo, surrogate modeling        |
| Spoken     | Catalan (native), Spanish (native), English (professional), Arabic (beginner) |

## Awards

|                |                                                                                                                                                                      |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nov 2024, 2025 | Participant, Datathon FME                                                                                                                                            |
| Jun 2025       | Physics in Ljubljana, summer school at the University of Ljubljana                                                                                                   |
| Since 2023     | Cum laude in several subjects                                                                                                                                        |
| 2023           | Admitted to CFIS on a grant, to read the two degrees at once. It takes a small cohort each year by its own entrance exam in mathematics and physics                  |
| 2019 - 2023    | Graduated from high school with honors, and represented it at the Barcelona <i>Mostra de Recerca Jove</i> , after UPC's summer courses in algorithms and programming |